Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2020 – Paper 1 – Solutions**

June 2020 – Multiple Choice Questions 1-15 – Solution

1. In the real number system the inverse of addition is represented by

   (A) $x + 0 = x$

   (B) $x + (-x) = 0$

   (C) $0 + x = x + 0$

   (D) $x(y + z) = xy + xz$

   The inverse of a number, x, is $(-x)$.

   The additive identity is 0.

   Hence,

   $x + (-x) = 0$

   Answer: B

2. Let $ax^3 + 6x^2 + cx + d = 0$. Given that $f(s) = 0$ and $f(-r) = 0$, which of the following equations is true?

(A) $(x - s)(x + r)(ax + p) = 0$

(B) $(x + s)(x - r)(ax + p) = 0$

(C) $(x - s)(x - r)(ax + p) = 0$

(D) $(x + s)(x + r)(ax + p) = 0$

Let $f(x) = ax^3 + 6x^2 + cx + d$.

Since $f(s) = 0$, then by the Factor Theorem, $(x - s)$ is a factor of $f(x)$.

Since $f(-r) = 0$, then by the Factor Theorem, $(x + r)$ is a factor of $f(x)$.

Answer: A

3. $\boldsymbol{p}$, $\boldsymbol{q}$ and $\boldsymbol{r}$ are the propositions such that

$\boldsymbol{p}$: it is raining

$\boldsymbol{q}$: the sun is shining

$\boldsymbol{r}$: there is a rainbow

The expression "it is raining and the sun is shining, but there is no rainbow" in symbolic form is

(A) $(\boldsymbol{p} \vee \boldsymbol{q}) \vee \sim\boldsymbol{r}$

(B) $(\boldsymbol{p} \vee \boldsymbol{q}) \wedge \sim\boldsymbol{r}$

(C) $(\boldsymbol{p} \wedge \boldsymbol{q}) \wedge \sim\boldsymbol{r}$

(D) $(\boldsymbol{p} \wedge \boldsymbol{q}) \vee \sim\boldsymbol{r}$

Expression: "it is raining and the sun is shining, but there is no rainbow"

Symbolic form: $(\boldsymbol{p} \wedge \boldsymbol{q}) \wedge \sim\boldsymbol{r}$

Answer: C

4. If $\log_a 4 + \log_a x - \log_a 7 = 2$, then the value of x is

(A) $\frac{7}{2^a}$

(B) $\frac{7}{4}a^2$

(C) $\frac{4}{7}a^2$

(D) $\frac{7}{4}2a$

$\log_a 4 + \log_a x - \log_a 7 = 2$

$$\log_a \frac{4x}{7} = 2$$

Converting to exponential form gives:

$\frac{4x}{7} = a^2$

$4x = 7a^2$

$x = \frac{7}{4}a^2$

Answer: B

5. $4\sqrt{x} + \frac{4\sqrt{3x}}{\sqrt{48}} =$

(A) $5\sqrt{x}$

(B) $\frac{\sqrt{x}}{3}$

(C) $4\sqrt{x} + 1$

(D) $4\sqrt{x} + \frac{\sqrt{x}}{4}$

$$4\sqrt{x} + \frac{4\sqrt{3x}}{\sqrt{48}} = 4\sqrt{x} + \frac{4\sqrt{3}\sqrt{x}}{\sqrt{16}\sqrt{3}}$$

$$= 4\sqrt{x} + \frac{4\sqrt{x}}{4}$$

$$= 4\sqrt{x} + \sqrt{x}$$

$$= 5\sqrt{x}$$

Answer: A

6. Given that x and y are negative integers, and that $x > y$, which of the following is true?

(A) $x^2 > y^2$

(B) $x^2 < y$

(C) $x > y^2$

(D) $x^2 < y^2$

Consider $x = -2$ and $y = -3$.

Then $-2 > -3$ where $x > y$.

Squaring both sides gives,

$(-2)^2 = 4$ and $(-3)^2 = 9$

Since $4 < 9$, then $x^2 < y^2$.

Answer: D

7. The values of x that satisfy the inequality $|2x - a| > |x|$, $a > 0$, are

(A) $x < \frac{a}{3}$ or $x > a$

(B) $x < \frac{-a}{3}$ or $x > a$

(C) $x > -a$ and $x < \frac{a}{3}$

(D) $x < a$ and $x > \frac{a}{3}$

$|2x - a| > |x|$

Squaring both sides gives,

$|2x - a|^2 > |x|^2$

Using the property, $|x|^2 = x^2$, we have,

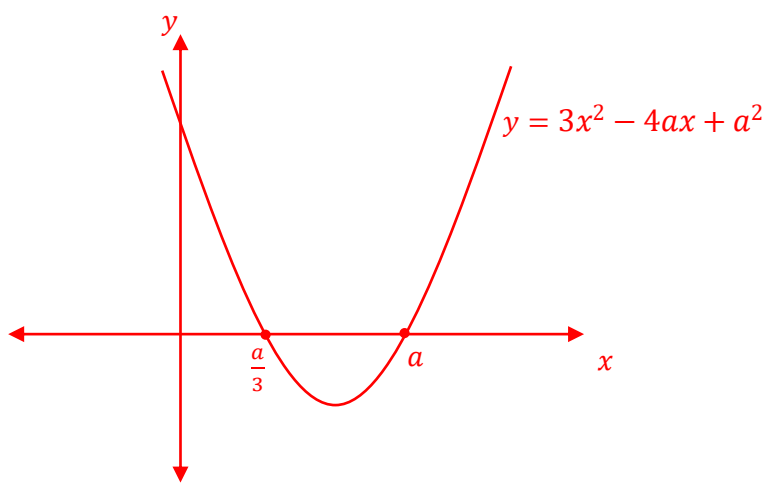
$$(2x - a)^2 > x^2$$

$$4x^2 - 4ax + a^2 > x^2$$

$$3x^2 - 4ax + a^2 > 0$$

$$(3x - a)(x - a) > 0$$

The critical values are $x = \frac{a}{3}$ and $x = a$.



Since we are solving $3x^2 - 4ax + a^2 > 0$, we look where the graph is positive.

The graph is positive for $x < \frac{a}{3}$ or $x > a$.

Answer: A

8. $P(x) = x^2 - hx - 12h^2$ and $P(\alpha) = 0$. What are the values of h, in terms of α?

(A) $-\frac{\alpha}{3}, \frac{\alpha}{4}$

(B) $-3\alpha, 4\alpha$

(C) $\frac{\alpha}{3}, \frac{\alpha}{4}$

(D) $\alpha, 4\alpha$

$P(x) = x^2 - hx - 12h^2$

When $P(\alpha) = 0$,

$0 = \alpha^2 - \alpha h - 12h^2$

$0 = (\alpha - 4h)(\alpha + 3h)$

Either $\alpha - 4h = 0$ or $\alpha + 3h = 0$

$4h = \alpha$ $\quad$ $3h = -\alpha$

$h = \frac{\alpha}{4}$ $\quad$ $h = -\frac{\alpha}{3}$

Answer: A

9. If the range of $f(x) = \frac{2}{x} + 1$ is $\left\{\frac{3}{2}, 2, 3, 5\right\}$, the domain is

(A) $\left\{\frac{1}{4}, \frac{1}{2}, 1, 2\right\}$

(B) $\{2, 3, 5, 9\}$

(C) $\left\{\frac{1}{2}, 1, 2, 4\right\}$

(D) $\left\{\frac{2}{3}, 1, 2, 3\right\}$

$f(x) = \frac{2}{x} + 1$

When $f(x) = \frac{3}{2}$,

$\frac{3}{2} = \frac{2}{x} + 1$

$\frac{3}{2} - 1 = \frac{2}{x}$

$\frac{1}{2} = \frac{2}{x}$

$x = 2 \times 2$

$x = 4$

Option C is the only option whose set contains the element 4.

Answer: C

10. The sum of the squares of the roots of the equation $4x^3 + 3x^2 - 2x + 1 = 0$ is

(A) $\frac{-9}{16}$

(B) $\frac{-7}{16}$

(C) $\frac{9}{16}$

(D) $\frac{25}{16}$

$4x^3 + 3x^2 - 2x + 1 = 0$

$(\div 4)$

$x^3 + \frac{3}{4}x^2 - \frac{1}{2}x + \frac{1}{4} = 0$

$\alpha + \beta + \gamma = -\frac{3}{4}$

$\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{1}{2}$

Now,

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$$

$$= \left(-\frac{3}{4}\right)^2 - 2\left(-\frac{1}{2}\right)$$

$$= \frac{9}{16} + 1$$

$$= \frac{9}{16} + \frac{16}{16}$$

$$= \frac{25}{16}$$

Answer: D

11. Which of the following statements is true?

(A) $\sum_{r=1}^{n} r^2 = 2\sum_{r=1}^{n} r$

(B) $\sum_{r=1}^{n} r^2 = [\sum_{r=1}^{n} r]^2$

(C) $\sum_{r=1}^{n} r^2 = \sum_{r=1}^{n} r^2 - 1$

(D) $\sum_{r=1}^{n}(2 + r^2) = 2n + \sum_{r=1}^{n} r^2$

$$\sum_{r=1}^{n}(2 + r^2) = \sum_{r=1}^{n} 2 + \sum_{r=1}^{n} r^2$$

$$= 2n + \sum_{r=1}^{n} r^2$$

Answer: D

12. Which of the following are factors of $4x^4 + 8x^3 - 2x^2 - 6x - 4$?

I. $x + 1$

II. $x - 1$

III. $x + 2$

(A) I and II only

(B) I and III only

(C) II and III only

(D) I, II and III

Let $f(x) = 4x^4 + 8x^3 - 2x^2 - 6x - 4$.

Consider $(x + 1)$.

If $(x + 1)$ is a factor of $f(x)$, then by the Factor Theorem, $f(-1) = 0$.

$$f(-1) = 4(-1)^4 + 8(-1)^3 - 2(-1)^2 - 6(-1) - 4$$

$$= 4 - 8 - 2 + 6 - 4$$

$$= -4$$

So, $(x + 1)$ is not a factor of $f(x)$.

Consider $(x - 1)$.

If $(x - 1)$ is a factor of $f(x)$, then by the Factor Theorem, $f(1) = 0$.

$$f(1) = 4(1)^4 + 8(1)^3 - 2(1)^2 - 6(1) - 4$$

$$= 4 + 8 - 2 - 6 - 4$$

$$= 0$$

So, $(x - 1)$ is a factor of $f(x)$.

Consider $(x+2)$.

If $(x+2)$ is a factor of $f(x)$, then by the Factor Theorem, $f(-2)=0$.

$$f(-2) = 4(-2)^4 + 8(-2)^3 - 2(-2)^2 - 6(-2) - 4$$

$$= 4(16) + 8(-8) - 2(4) - 6(-2) - 4$$

$$= 64 - 64 - 8 + 12 - 4$$

$$= 0$$

So, $(x+2)$ is a factor of $f(x)$.

Answer: C

13. Two roots of the cubic equation $2x^3 + 3x^2 - 5x - 6 = 0$ are -1 and -2. The THIRD root is

(A) $\frac{-3}{2}$

(B) $\frac{1}{2}$

(C) $\frac{3}{2}$

(D) 3

Since $x = -1$ and $x = -2$ are two roots of the cubic equation, we can state that:

$2x^3 + 3x^2 - 5x - 6 = (x + 1)(x + 2)(ax + b)$

Since the coefficient of x^3 must be 2, then $a = 2$.

Since the constant is -6, then

$$1 \times 2 \times b = -6$$

$$2b = -6$$

$$b = -\frac{6}{2}$$

$$b = -3$$

So, the third factor is $(2x - 3)$. The third root is

$$2x - 3 = 0$$

$$2x = 3$$

$$x = \frac{3}{2}$$

Answer: C

14. $3\log_2 2q - 2\log_2 3q + 1$ expressed as a single logarithm in its SIMPLEST form is

(A) $\log_2\left(\frac{16q}{9}\right)$

(B) $\log_2\left(\frac{8q}{9}\right)$

(C) $\log_2\left(\frac{2q}{3}\right) + 1$

(D) $\frac{3}{2}\log_2\left(\frac{2q}{3}\right) + 1$

$$3\log_2 2q - 2\log_2 3q + 1 = \log_2(2q)^3 - \log_2(3q)^2 + \log_2 2$$

$$= \log_2\left(\frac{(2q)^3(2)}{(3q)^2}\right)$$

$$= \log_2\left(\frac{16q^3}{9q^2}\right)$$

$$= \log_2\left(\frac{16q}{9}\right)$$

Answer: A

15. The statement $\sim(\boldsymbol{p} \vee (\sim\boldsymbol{p} \wedge \boldsymbol{q}))$ is logically equivalent to

(A) $\boldsymbol{p} \wedge \sim\boldsymbol{q}$

(B) $\boldsymbol{p} \rightarrow \sim\boldsymbol{q}$

(C) $\sim\boldsymbol{p} \rightarrow \sim\boldsymbol{q}$

(D) $\sim\boldsymbol{p} \wedge \sim\boldsymbol{q}$

$\sim(\boldsymbol{p} \vee (\sim\boldsymbol{p} \wedge \boldsymbol{q})) = \sim[(\boldsymbol{p} \vee \sim\boldsymbol{p}) \wedge (\boldsymbol{p} \vee \boldsymbol{q})]$ [Distributive Law]

$= \sim[T \wedge (\boldsymbol{p} \vee \boldsymbol{q})]$ [Complement Law]

$= \sim(\boldsymbol{p} \vee \boldsymbol{q})$ [Identity Law]

$= \sim\boldsymbol{p} \wedge \sim\boldsymbol{q}$ [De Morgan's Law]

Answer: D

June 2020 – Multiple Choice Questions 16-30 – Solution

16. A circle is defined as $(x+2)^2 + y^2 = 4$. The gradient of the normal to the circle at $(0, 0)$ is

(A) ∞

(B) $\frac{1}{2}$

(C) 0

(D) -2

$(x+2)^2 + y^2 = 4$

The centre of the circle is $(-2, 0)$.

The points are $(-2, 0)$ and $(0, 0)$ which both lie on the x-axis.

The gradient of the horizontal line formed by these two points is $m = 0$.

Answer: C

17. The expression $\sin 6A + \sin 4A$ may be written as

(A) $\sin 10A$

(B) $-2\cos 2A$

(C) $2\cos 5A \sin A$

(D) $2\sin 5A \cos A$

$$\sin 6A + \sin 4A = 2\sin\left(\frac{6A+4A}{2}\right)\cos\left(\frac{6A-4A}{2}\right)$$

$$= 2\sin\left(\frac{10A}{2}\right)\cos\left(\frac{2A}{2}\right)$$

$$= 2\sin 5A \cos A$$

Answer: D

18. $8\sin\theta - 5\cos\theta$ has a MINIMUM value of

(A) -89

(B) -13

(C) $-\sqrt{89}$

(D) $\sqrt{89}$

Let $8\sin\theta - 5\cos\theta = r\sin(\theta - \alpha)$

$r = \sqrt{(8)^2 + (5)^2}$

$= \sqrt{64 + 25}$

$= \sqrt{89}$

Now, $-\sqrt{89} \leq r\sin(\theta - \alpha) \leq \sqrt{89}$

Answer: C

19. The relationship between the curve $y = x^2 - 2x + 4$ and the line $y = 2x$ is that the line

(A) does not intersect the curve

(B) intersects the curve in more than 2 points

(C) touches the curve at the point $(2, 4)$

(D) intersects the curve in two distinct points $(2, 2)$ and $(2, -2)$

$y = x^2 - 2x + 4 \quad \rightarrow$ Equation 1

$y = 2x \quad \rightarrow$ Equation 2

Equating both equations gives:

$$2x = x^2 - 2x + 4$$

$$x^2 - 2x - 2x + 4 = 0$$

$$x^2 - 4x + 4 = 0$$

which is in the form $ax^2 + bx + c = 0$

where $a = 1, b = -4$ and $c = 4$.

Now,

$$b^2 - 4ac = (-4)^2 - 4(1)(4)$$

$$= 16 - 16$$

$$= 0$$

Since $b^2 - 4ac = 0$, then the line touches the curve at one point.

$$x^2 - 4x + 4 = 0$$

$$(x-2)^2 = 0$$

$$x - 2 = 0$$

$$x = 2$$

When $x = 2$,

$y = 2(2)$

$= 4$

$\therefore$ The line touches the curve at the point $(2, 4)$.

Answer: C

20. A curve is defined by the parametric equations $x = 3 + 2t$ and $y = 2 + t$. The Cartesian equation of the curve is

(A) $x + y = 1$

(B) $x + 4y = 7$

(C) $x + 2y = 1$

(D) $x - 2y = -1$

$x = 3 + 2t$ → Equation 1

$y = 2 + t$ → Equation 2

Rearranging Equation 2 gives:

$t = y - 2$ → Equation 3

Substituting Equation 3 into Eqation 1 gives:

$x = 3 + 2(y - 2)$

$x = 3 + 2y - 4$

$x = 2y - 1$

$x - 2y = -1$

Answer: D

21. Given that the position vectors of the points P and Q are $\begin{pmatrix} 2 \\ -1 \\ 8 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 4 \\ 4 \end{pmatrix}$,

respectively, then the length of the vector $\overrightarrow{PQ}$ is

(A) $3\sqrt{2}$

(B) $5\sqrt{2}$

(C) 4

(D) 3

We are given that $\overrightarrow{OP} = \begin{pmatrix} 2 \\ -1 \\ 8 \end{pmatrix}$ and $\overrightarrow{OQ} = \begin{pmatrix} -1 \\ 4 \\ 4 \end{pmatrix}$.

Using the triangle law,

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$$

$$= \begin{pmatrix} -1 \\ 4 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 8 \end{pmatrix}$$

$$= \begin{pmatrix} -1-2 \\ 4-(-1) \\ 4-8 \end{pmatrix}$$

$$= \begin{pmatrix} -3 \\ 5 \\ -4 \end{pmatrix}$$

Now,

$$|\overrightarrow{PQ}| = \sqrt{(-3)^2 + (5)^2 + (-4)^2}$$

$$= \sqrt{9 + 25 + 16}$$

$$= \sqrt{50}$$

$= \sqrt{25 \times 2}$

$= \sqrt{25} \times \sqrt{2}$

$= 5\sqrt{2}$ units

Answer: B

22. The expression $\cot x + \tan x$ can be written as

(A) $2 \operatorname{cosec} 2x$

(B) $2 \cot 2x$

(C) $\frac{\sin x + \cos x}{\sin x \cos x}$

(D) $\frac{\sin x - \cos x}{\sin x \cos x}$

$$\cot x + \tan x = \frac{\cos x}{\sin x} + \frac{\sin x}{\cos x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\sin x \cos x}$$

$$= \frac{1}{\sin x \cos x}$$

$$= \frac{2}{2 \sin x \cos x}$$

$$= \frac{2}{\sin 2x}$$

$$= 2 \operatorname{cosec} 2x$$

Answer: A

23. The general solution for $\sin 2\theta = \sin\frac{\pi}{6}$, where $n \in \mathbb{Z}$, is

(A) $\theta = \begin{cases} 2n\pi + \frac{\pi}{6} \\ (2n+1)\frac{5\pi}{16} \end{cases}$

(B) $\theta = \begin{cases} n\pi + \frac{\pi}{12} \\ n\pi + \frac{5\pi}{12} \end{cases}$

(C) $\theta = \begin{cases} n\pi + \frac{\pi}{12} \\ (2n\pi)\frac{5\pi}{12} \end{cases}$

(D) $\theta = \begin{cases} n\pi + \frac{\pi}{6} \\ (n+1)\frac{5\pi}{6} \end{cases}$

$\sin 2\theta = \sin\frac{\pi}{6}$

$PV = \frac{\pi}{6}$

G.S. is $2\theta = n\pi + (-1)^n\left(\frac{\pi}{6}\right)$ $\quad , n \in \mathbb{Z}$

$\theta = \frac{n\pi}{2} + (-1)^n\left(\frac{\pi}{12}\right)$ $\quad , n \in \mathbb{Z}$

Consider when n is even.

When $n = 2n$,

$\theta = \frac{2n\pi}{2} + (-1)^{2n}\left(\frac{\pi}{12}\right)$

$= n\pi + \left(\frac{\pi}{12}\right)$

Consider when n is odd.

When $n = 2n + 1$,

$$\theta = \frac{(2n+1)\pi}{2} + (-1)^{2n+1}\left(\frac{\pi}{12}\right)$$

$$= \frac{2n\pi}{2} + \frac{\pi}{2} - \frac{\pi}{12}$$

$$= n\pi + \frac{5\pi}{12}$$

$\therefore$ The G.S. is $\theta = \begin{cases} n\pi + \frac{\pi}{12} \\ n\pi + \frac{5\pi}{12} \end{cases}$.

Answer: B

24. The vector $\boldsymbol{u}$ has magnitude $4\sqrt{5}$ units and is parallel to the vector $\boldsymbol{v} = \hat{\boldsymbol{\imath}} - 2\hat{\boldsymbol{\jmath}}$. A unit vector parallel to $\boldsymbol{u}$ is

(A) $\frac{\sqrt{5}}{20}(\hat{\boldsymbol{\imath}} - 2\hat{\boldsymbol{\jmath}})$

(B) $\frac{4}{\sqrt{5}}(\hat{\boldsymbol{\imath}} - 2\hat{\boldsymbol{\jmath}})$

(C) $4\sqrt{5}(\hat{\boldsymbol{\imath}} - 2\hat{\boldsymbol{\jmath}})$

(D) $\frac{1}{\sqrt{5}}(\hat{\boldsymbol{\imath}} - 2\hat{\boldsymbol{\jmath}})$

$\boldsymbol{v} = \hat{\boldsymbol{\imath}} - 2\hat{\boldsymbol{\jmath}}$

$\boldsymbol{u} = a(\hat{\boldsymbol{\imath}} - 2\hat{\boldsymbol{\jmath}})$

$= a\hat{\boldsymbol{\imath}} - 2a\hat{\boldsymbol{\jmath}}$

$|\boldsymbol{u}| = \sqrt{a^2 + (-2a)^2}$

$= \sqrt{a^2 + 4a^2}$

$= \sqrt{5a^2}$

$= \sqrt{5}a$

Since $|\boldsymbol{u}| = 4\sqrt{5}$ units, then $a = 4$.

So, we have

$\boldsymbol{u} = 4\hat{\boldsymbol{\imath}} - 2(4)\hat{\boldsymbol{\jmath}}$

$= 4\hat{\boldsymbol{\imath}} - 8\hat{\boldsymbol{\jmath}}$

Now,

A unit vector parallel to $\boldsymbol{u}$ is

$$\hat{\boldsymbol{u}} = \frac{\boldsymbol{u}}{|\boldsymbol{u}|}$$

$$= \frac{4\hat{\boldsymbol{i}} - 8\hat{\boldsymbol{j}}}{4\sqrt{5}}$$

$$= \frac{4(\hat{\boldsymbol{i}} - 2\hat{\boldsymbol{j}})}{4\sqrt{5}}$$

$$= \frac{(\hat{\boldsymbol{i}} - 2\hat{\boldsymbol{j}})}{\sqrt{5}}$$

$$= \frac{1}{\sqrt{5}}(\hat{\boldsymbol{i}} - 2\hat{\boldsymbol{j}})$$

Answer: D

25. The variable point $P(x, y)$ moves so that it is the same distance from the points $(1, 6)$ and $(3, 2)$. The equation of the locus of P may be obtained from

(A) $\frac{y-6}{x-1} = \frac{y-2}{x-3}$

(B) $\frac{x-1}{y-6} = \frac{x-3}{y-2}$

(C) $\sqrt{(x-1)^2 + (y-6)^2} = \sqrt{(x-3)^2 + (y-2)^2}$

(D) $(x-1) + (y-6) = (x-3) + (y-2)$

Consider the points $P(x, y)$ and $(1, 6)$.

Distance $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{(x-1)^2 + (y-6)^2}$

Consider the points $P(x, y)$ and $(3, 2)$.

Distance $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{(x-3)^2 + (y-2)^2}$

Now,

The equation of the locus of P is

$\sqrt{(x-1)^2 + (y-6)^2} = \sqrt{(x-3)^2 + (y-2)^2}$

Answer: C

26. The circle with equation $x^2 + y^2 - 2x - 6y + 1 = 0$ cuts the circle with equation $x^2 + y^2 - 8x - 8y + 31 = 0$ in two distinct places. The equation of the common chord is

(A) $x - y + 15 = 0$

(B) $x + 3y - 15 = 0$

(C) $3x + y - 15 = 0$

(D) $3y + x + 15 = 0$

$x^2 + y^2 - 2x - 6y + 1 = 0$ → Equation 1

$x^2 + y^2 - 8x - 8y + 31 = 0$ → Equation 2

Equating both equations gives:

$$x^2 + y^2 - 2x - 6y + 1 = x^2 + y^2 - 8x - 8y + 31$$

$$-2x - 6y + 1 = -8x - 8y + 31$$

$$-2x - 6y + 1 + 8x + 8y - 31 = 0$$

$$6x + 2y - 30 = 0$$

$$(\div 2)$$

$$3x + y - 15 = 0$$

∴ The equation of the common chord is $3x + y - 15 = 0$.

Answer: C

27. The cosine of the angle between the vectors $-6\hat{\boldsymbol{j}}$ and $\hat{\boldsymbol{i}} + \hat{\boldsymbol{j}}$ is

(A) $\frac{-1}{\sqrt{2}}$

(B) $\frac{1}{\sqrt{2}}$

(C) $\frac{-5}{\sqrt{2}}$

(D) $\frac{6}{\sqrt{2}}$

The vectors are $-6\hat{\boldsymbol{j}}$ and $\hat{\boldsymbol{i}} + \hat{\boldsymbol{j}}$ which can be written as $\boldsymbol{a} = \begin{pmatrix} 0 \\ -6 \end{pmatrix}$ and $\boldsymbol{b} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

$$\boldsymbol{a} \cdot \boldsymbol{b} = \begin{pmatrix} 0 \\ -6 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= (0)(1) + (-6)(1)$$

$$= 0 - 6$$

$$= -6$$

Also,

$$|\boldsymbol{a}| = \sqrt{(0)^2 + (-6)^2}$$

$$= \sqrt{0 + 36}$$

$$= \sqrt{36}$$

$$= 6 \text{ units}$$

$$|\boldsymbol{b}| = \sqrt{(1)^2 + (1)^2}$$

$$= \sqrt{1 + 1}$$

$$= \sqrt{2}$$

Now,

$$\cos\theta = \frac{\boldsymbol{a}\cdot\boldsymbol{b}}{|\boldsymbol{a}||\boldsymbol{b}|}$$

$$= \frac{-6}{6\sqrt{2}}$$

$$= \frac{-1}{\sqrt{2}}$$

Answer: A

28. The set of values of k for which the straight line $y = kx - 4$ intersects the curve with equation $y = x^2 - 2x$ at two distinct points is

(A) $\{k < -1\} \cup \{k > 6\}$

(B) $\{-2 < k < 6\}$

(C) $\{k < -6\} \cup \{k > 2\}$

(D) $\{-6 < k < 2\}$

$y = kx - 4$ → Equation 1

$y = x^2 - 2x$ → Equation 2

Equating both equations gives:

$$kx - 4 = x^2 - 2x$$

$$x^2 - 2x - kx + 4 = 0$$

$$x^2 + (-2 - k)x + 4 = 0$$

which is in the form $ax^2 + bx + c = 0$,

where $a = 1, b = -2 - k$ and $c = 4$.

In order for the line to intersect the curve at two distinct points,

$$b^2 - 4ac > 0$$

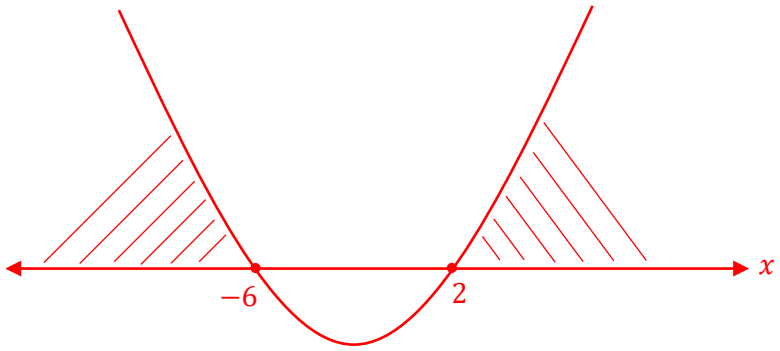
$$(-2 - k)^2 - 4(1)(4) > 0$$

$$4 + 4k + k^2 - 16 > 0$$

$$k^2 + 4k - 12 > 0$$

$$(k + 6)(k - 2) > 0$$

The critical values are $k = -6$ and $k = 2$.



Since we are solving $(k+6)(k-2) > 0$, we look where the graph is positive.

The graph is positive for $k < -6$ and $k > 2$.

$\therefore$The solution set is $\{k < -6\} \cup \{k > 2\}$.

Answer: C

29. Given that $\boldsymbol{a} = \begin{pmatrix} -3 \\ 7 \end{pmatrix}$ and $\boldsymbol{b} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$, then $|3\boldsymbol{a} + 2\boldsymbol{b}|$ is equal to

(A) $\sqrt{336}$

(B) $\sqrt{386}$

(C) $\sqrt{504}$

(D) $\sqrt{554}$

$$3\boldsymbol{a} + 2\boldsymbol{b} = 3\begin{pmatrix} -3 \\ 7 \end{pmatrix} + 2\begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} -9 \\ 21 \end{pmatrix} + \begin{pmatrix} 4 \\ -2 \end{pmatrix}$$

$$= \begin{pmatrix} -9 + 4 \\ 21 + (-2) \end{pmatrix}$$

$$= \begin{pmatrix} -5 \\ 19 \end{pmatrix}$$

Now,

$$|3\boldsymbol{a} + 2\boldsymbol{b}| = \sqrt{(-5)^2 + (19)^2}$$

$$= \sqrt{25 + 361}$$

$$= \sqrt{386}$$

Answer: B

30. The path of an object is given parametrically as $x = \sin t + 2$, $y = \cos t + 1$. The slope of the tangent at $t = \frac{\pi}{4}$ is

(A) 0

(B) -1

(C) 1

(D) undefined

Module 2 Approach:

$x = \sin t + 2$, $y = \cos t + 1$

$x - 2 = \sin t$ $y - 1 = \cos t$

$(x-2)^2 = \sin^2 t$ $(y-1)^2 = \cos^2 t$

Now,

$\sin^2 t + \cos^2 t = 1$

$(x-2)^2 + (y-1)^2 = 1$

which is in the form $(x-a)^2 + (y-b)^2 = r^2$,

where $C(2, 1)$ and radius, $r = 1$.

$x = \sin t + 2$, $y = \cos t + 1$

Substituting $t = \frac{\pi}{4}$ gives:

$x = \sin\frac{\pi}{4} + 2$ $y = \cos\frac{\pi}{4} + 1$

$= \frac{1}{\sqrt{2}} + 2$ $= \frac{1}{\sqrt{2}} + 1$

The points are $(2, 1)$ and $\left(\frac{1}{\sqrt{2}} + 2, \frac{1}{\sqrt{2}} + 1\right)$.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{\frac{1}{\sqrt{2}} + 1 - 1}{\frac{1}{\sqrt{2}} + 2 - 2}$$

$$= \frac{\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}$$

$$= 1$$

Therefore,

Gradient of normal $= 1$

Gradient of tangent $= -1$

Module 3 Approach:

$x = \sin t + 2$, $y = \cos t + 1$

$\frac{dx}{dt} = \cos t$ $\frac{dy}{dt} = -\sin t$

Using the chain rule,

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= -\sin t \times \frac{1}{\cos t}$$

$$= -\frac{\sin t}{\cos t}$$

Substituting $t = \frac{\pi}{4}$ gives,

$$\frac{dy}{dx} = -\frac{\sin\left(\frac{\pi}{4}\right)}{\cos\left(\frac{\pi}{4}\right)}$$

$$= -\frac{\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}}$$

$$= -1$$

Answer: B

June 2020 – Multiple Choice Questions 31-45 – Solution

31. Given that $y = (2x-1)^4$, then $\frac{d^2y}{dx^2} =$

(A) $8\sqrt{y}$

(B) $16\sqrt{y}$

(C) $24\sqrt{y}$

(D) $48\sqrt{y}$

$y = (2x-1)^4$

$\frac{dy}{dx} = 4(2)(2x-1)^{4-1}$

$= 8(2x-1)^3$

Now,

$\frac{d^2y}{dx^2} = 8(3)(2)(2x-1)^{3-1}$

$= 48(2x-1)^2$

$= 48[(2x-1)^4]^{\frac{1}{2}}$

$= 48y^{\frac{1}{2}}$

$= 48\sqrt{y}$

Answer: D

32. $\lim_{x \to -1} \frac{x^2-x-2}{x-1} =$

(A) -3

(B) 0

(C) 1

(D) ∞

$$\lim_{x \to -1} \frac{x^2-x-2}{x-1} = \frac{(-1)^2-(-1)-2}{(-1)-1}$$

$$= \frac{1+1-2}{-2}$$

$$= \frac{0}{-2}$$

$$= 0$$

Answer: B

33. Which of the following real functions has a point of discontinuity?

(A) $f(x) = \sqrt{x^2 + 9}$

(B) $f(x) = \frac{x}{x^2+1}$

(C) $f(x) = 3 - x^2$

(D) $f(x) = \frac{4}{(x+1)^2}$

A function has a point of discontinuity when the denominator is zero.

Consider option D.

$f(x) = \frac{4}{(x+1)^2}$

When the denominator is equal to 0,

$(x + 1)^2 = 0$

$x + 1 = 0$

$x = -1$

$\therefore f(x) = \frac{4}{(x+1)^2}$ has a point of discontinuity when $x = -1$.

Answer: D

34. If the point $A(3,-1,2)$ lies on the plane $x-2y+3z=d$, then the value of the constant d is

(A) 7

(B) 11

(C) -7

(D) -11

Let $r=\begin{pmatrix}x\\y\\z\end{pmatrix}$, $a=\begin{pmatrix}3\\-1\\2\end{pmatrix}$ and $n=\begin{pmatrix}1\\-2\\3\end{pmatrix}$.

Now,

$$\boldsymbol{r}\cdot\boldsymbol{n}=\boldsymbol{a}\cdot\boldsymbol{n}$$

$$\begin{pmatrix}x\\y\\z\end{pmatrix}\cdot\begin{pmatrix}1\\-2\\3\end{pmatrix}=\begin{pmatrix}3\\-1\\2\end{pmatrix}\cdot\begin{pmatrix}1\\-2\\3\end{pmatrix}$$

$$(x)(1)+(y)(-2)+(z)(3)=(3)(1)+(-1)(-2)+(2)(3)$$

$$x-2y+3z=3+2+6$$

$$x-2y+3z=11$$

$\therefore d=11$

Answer: B

35. If $y = \frac{x-6}{3-4x}$ then $\frac{dy}{dx}$ is

(A) $\frac{-21}{(3-4x)^2}$

(B) $\frac{21}{(3-4x)^2}$

(C) $\frac{27-8x}{(3-4x)^2}$

(D) $\frac{-27-8x}{(3-4x)^2}$

$y = \frac{x-6}{3-4x}$

which is in the form $y = \frac{u}{v}$.

Let $u = x - 6$ and $v = 3 - 4x$

$\frac{du}{dx} = 1$ $\frac{dv}{dx} = -4$

Using the quotient rule,

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

$$= \frac{(3-4x)(1)-(x-6)(-4)}{(3-4x)^2}$$

$$= \frac{3-4x+4x-24}{(3-4x)^2}$$

$$= \frac{-21}{(3-4x)^2}$$

Answer: A

36. The volume of the solid generated when the area enclosed by the curve $y = x^2$, the y-axis and the line $y = 4$ is rotated 360° about the y-axis is

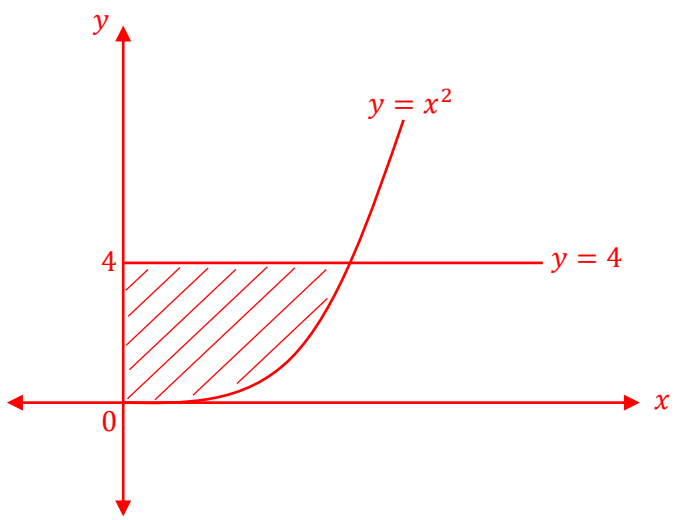
(A) 8π

(B) 16π

(C) 32π

(D) 64π

Consider the diagram below.



We have $y = x^2$.

Now,

The volume of the enclosed region is

$V = \pi \int_a^b x^2 \, dy$

$= \pi \int_0^4 y \, dy$

$= \pi \left[\frac{y^2}{2}\right]_0^4$

$= \frac{\pi}{2}[(4)^2 - (0)^2]$

$= \frac{\pi}{2}(16)$

$= 8\pi\ units^3$

Answer: A

37. A curve is defined by the equation $y = -5(1-2x)^2$. Given that x increases at a rate of 1 unit per second when $x = 1$, what is the corresponding rate of change for y?

(A) -40

(B) -20

(C) 20

(D) 40

$y = -5(1-2x)^2$

$\frac{dy}{dx} = -5(2)(-2)(1-2x)$

$= 20(1-2x)$

When $x = 1$,

$\frac{dy}{dx} = 20[1-2(1)]$

$= 20(-1)$

$= -20$

Using chain rule,

$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$

$= -20 \times 1$

$= -20$

Answer: B

38. A curve C has equation $y = 16x + \frac{1}{x}$. C has stationary points at $x =$

(A) $-\frac{1}{4}$

(B) $\frac{1}{4}$

(C) ± 4

(D) $\pm\frac{1}{4}$

$y = 16x + \frac{1}{x}$

$y = 16x + x^{-1}$

$\frac{dy}{dx} = 16 - x^{-2}$

At stationary points, $\frac{dy}{dx} = 0$.

When $\frac{dy}{dx} = 0$,

$16 - x^{-2} = 0$

$x^{-2} = 16$

$\frac{1}{x^2} = 16$

$x^2 = \frac{1}{16}$

$x = \pm\sqrt{\frac{1}{16}}$

$x = \pm\frac{1}{4}$

Answer: D

39. $\frac{d}{dx}(x^3 \sin x)$ may be expressed as

(A) $x^2(\cos x + 3\sin x)$

(B) $x^2(3\cos x + \sin x)$

(C) $x^2(x\cos x + 3\sin x)$

(D) $x^2(x\cos x - 3\sin x)$

Let $y = x^3 \sin x$

which is in the form $y = uv$.

Let $u = x^3$ and $v = \sin x$

$\frac{du}{dx} = 3x^2$ $\frac{dv}{dx} = \cos x$

Using the product rule,

$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$

$= x^3 \cos x + \sin x\,(3x^2)$

$= x^2(x\cos x + 3\sin x)$

Answer: C

40. Given that $\cos 2x = 2\cos^2 x - 1$, then $\int \cos^2\left(\frac{x}{4}\right)\, dx$ is

(A) $\frac{1}{2}x + \frac{1}{2}\int \cos 4x\ dx$

(B) $\frac{1}{2}x + \int \cos\frac{x}{4}\, dx$

(C) $\frac{1}{2}x + \int \cos\frac{x}{2}\, dx$

(D) $\frac{1}{2}x + \frac{1}{2}\int \cos\frac{x}{2}\, dx$

$$\cos 2x = 2\cos^2 x - 1$$

$$2\cos^2 x = \cos 2x + 1$$

$$\cos^2 x = \frac{1}{2}\cos 2x + \frac{1}{2}$$

Let $x = \frac{x}{4}$.

$$\cos^2\left(\frac{x}{4}\right) = \frac{1}{2}\cos 2\left(\frac{x}{4}\right) + \frac{1}{2}$$

$$\cos^2\left(\frac{x}{4}\right) = \frac{1}{2}\cos\frac{x}{2} + \frac{1}{2}$$

Integrating both sides gives:

$$\int \cos^2\left(\frac{x}{4}\right)\, dx = \int \left(\frac{1}{2}\cos\frac{x}{2} + \frac{1}{2}\right)\, dx$$

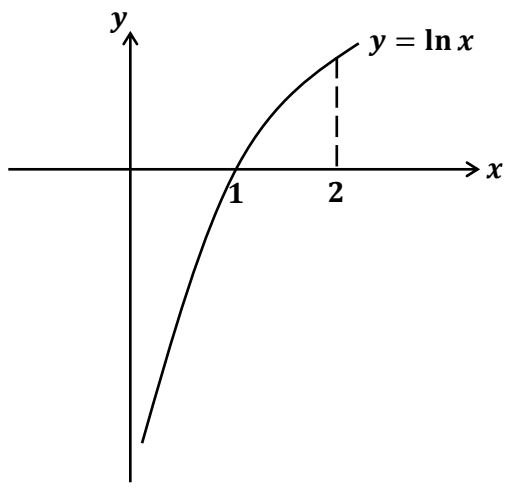
$$= \int \frac{1}{2}\cos\frac{x}{2}\, dx + \int \frac{1}{2}\, dx$$

$$= \frac{1}{2}\int \cos\frac{x}{2}\, dx + \frac{1}{2}x$$

$$= \frac{1}{2}x + \frac{1}{2}\int \cos\frac{x}{2}\, dx$$

Answer: D

Item **41** refers to the following diagram.



41. The gradient of the **normal** to the curve $y = \ln x$ at $x = 2$ is

(A) -2

(B) $-\frac{1}{2}$

(C) $\frac{1}{2}$

(D) 2

$y = \ln x$

$\frac{dy}{dx} = \frac{1}{x}$

When $x = 2$,

$\frac{dy}{dx} = \frac{1}{2}$

Gradient of the tangent $= \frac{1}{2}$

Gradient of the normal $= -2$

Answer: A

42. The radius of a circle is increasing at a rate of 0.1 cms^{-1}. At the instant when the radius is 3 cm, the rate of increase of the area, in cm^2s^{-1}, is

(A) $\frac{2}{5}\pi$

(B) $\frac{3}{5}\pi$

(C) 2π

(D) 4π

$A = \pi r^2$, $\frac{dr}{dt} = 0.1\ cms^{-1}$

$\frac{dA}{dr} = 2\pi r$

When $r = 3$,

$\frac{dA}{dr} = 2\pi(3)$

$= 6\pi$

Using the chain rule,

$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$

$= 6\pi \times 0.1$

$= 0.6\pi$

$= \frac{3}{5}\pi$

Answer: B

43. If the rate of change of y with respect to x is $3x^2 + \frac{4}{x^3}$, then y is equal to

(A) $6x - \frac{12}{x^4}$

(B) $x^3 + \frac{2}{x^4} + \text{constant}$

(C) $x^3 - \frac{2}{x^2} + \text{constant}$

(D) $6x - \frac{12}{x^2}$

$$\frac{dy}{dx} = 3x^2 + \frac{4}{x^3}$$

$$= 3x^2 + 4x^{-3}$$

Now,

$$y = \int \frac{dy}{dx} dx$$

$$= \int (3x^2 + 4x^{-3})\, dx$$

$$= \frac{3x^3}{3} + \frac{4x^{-2}}{-2} + c$$

$$= x^3 - \frac{2}{x^2} + c$$

Answer: C

44. $\lim_{x \to 2} \frac{x^3 - 2^3}{x - 2} =$

(A) 0

(B) 4

(C) 8

(D) 12

$$\lim_{x \to 2} \frac{x^3 - 2^3}{x - 2} = \lim_{x \to 2} \frac{(x-2)(x^2+2x+4)}{(x-2)}$$

$$= \lim_{x \to 2} (x^2 + 2x + 4)$$

$$= (2)^2 + 2(2) + 4$$

$$= 4 + 4 + 4$$

$$= 12$$

Answer: D

45. The gradient of the **normal** to the curve $y = 3x^2 - 2x + 1$ at $x = 1$ is

(A) $-\frac{1}{2}$

(B) $-\frac{1}{4}$

(C) -1

(D) 4

$y = 3x^2 - 2x + 1$

$\frac{dy}{dx} = 6x - 2$

When $x = 1$,

$\frac{dy}{dx} = 6(1) - 2$

$= 6 - 2$

$= 4$

Gradient of tangent $= 4$

Gradient of normal $= -\frac{1}{4}$

Answer: B